

Question	Answer	Marks
2(e)	- The (resultant) force is constant / does not decrease - air resistance would vary / decrease / not constant - the force is not zero when speed / velocity is zero / at 8 s Any two of the above 3 marking points (1 mark each, max 2)	B2
2(f)	line from origin with decreasing positive gradient	B1
	gradient changes from positive to negative at 8.0 s	B1
	after $t = 8.0$ s the line has a negative gradient of increasing magnitude <u>and</u> a positive value of d at $t = 12$ s	B1

Question	Answer	Marks
3(a)	$E_{(K)} = \frac{1}{2}mv^2$ $110 = \frac{1}{2} \times 5.5 \times v^2$	C1
	$v = 6.3 \text{ m s}^{-1}$	A1
3(b)	$(\Delta)E_{(P)} / 20 = mg(\Delta)h$ OR $(\Delta)E_{(P)} / 20 = mgx_0$	C1
	$(x_0 =) 20 / 5.5 \times 9.81 = 0.37 \text{ (m)}$ Allow $(\Delta)h$ for x_0	A1
3(c)	$[\max (E_{(P)})] = 110 + 20 = 130 \text{ J}$	A1